



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



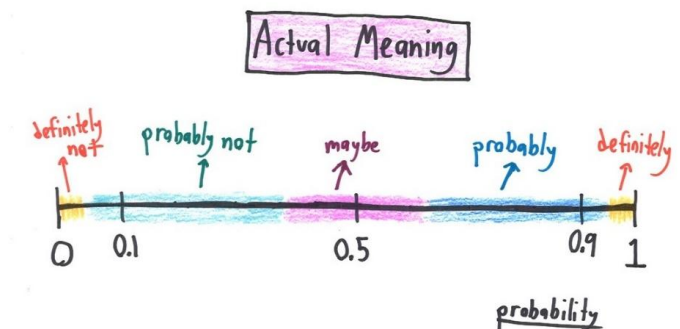


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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Week 8 - Lecture No 15 – October 08, 2025



Announcements!

1. Lectures up to No 13 posted. All quiz scores posted.
2. Midterm Exam - October 8, 2025, from 05:40 PM to 06:55 PM – **venue H-507 (L2)** – Good Luck

Agenda for today

1. Quiz No 04
2. Unit 4 - Discrete Random Variables

Unit 4
Discrete Random Variables

Multiple random variables and joint PMFs

- $X : p_X$
 - $Y : p_Y$
- marginal pmfs*
- $P(X = Y) = 2/20$

Joint PMF: $p_{X,Y}(x, y) = P(X = x \text{ and } Y = y)$

Y =
Second roll

4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = First roll $P_X(3)$

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

$$P_{X,Y}(1, 3) = 2/20$$

$$P_X(4) = 1/20 + 2/20$$

$$P_Y(2) = (1 + 3 + 1)/20$$

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$



Exercise No 01!

Suppose you roll two fair, six-sided dice, one of which is **red** and the other of which is **green**. Define the following two random variables **X** & **Y**:

X = The number shown on the **red die**

Y =
$$\begin{cases} 0 & \text{if the two dice show the same number} \\ 1 & \text{if the number on the green die is bigger than the number on the red die} \\ 2 & \text{if the number on the red die is bigger than the number on the green die} \end{cases}$$

Example



X =
Y =

1. Write down a table showing the joint probability mass function for **X** and **Y**.

		X =					
		1	2	3	4	5	6
Y =	0						
	1						
	2						



Exercise No 01!

$X =$

		1	2	3	4	5	6
	0	1/36	1/36	1/36	1/36	1/36	1/36
$Y =$	1	5/36	4/36	3/36	2/36	1/36	0
	2	0	1/36	2/36	3/36	4/36	5/36

2. Find the marginal probability mass function for Y , and compute its expected value.

y	$P(Y = y)$
0	
1	
2	

$$E(Y) = 0 \times (6/36) + 1 \times (15/36) + 2 \times (15/36) = 45/36$$



Exercise No 01!

$X =$

		1	2	3	4	5	6
$Y =$	0	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
	1	$5/36$	$4/36$	$3/36$	$2/36$	$1/36$	0
	2	0	$1/36$	$2/36$	$3/36$	$4/36$	$5/36$

3. Find the conditional probability mass function for X given $Y = 1$.

$X =$

1	2	3	4	5	6
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More than two random variables

$$p_{X,Y,Z}(x, y, z) = \mathbf{P} (X = x \text{ and } Y = y \text{ and } Z = z)$$

$$\rightarrow \sum_x \sum_y \sum_z p_{X,Y,Z}(x, y, z) = 1$$

$$\rightarrow p_X(x) = \sum_y \sum_z p_{X,Y,Z}(x, y, z)$$

$$\rightarrow p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z)$$

Functions of multiple random variables

$$Z = g(X, Y)$$

$$\rightarrow p_Z(z) = \mathbf{P}(Z = z) = \mathbf{P}(g(X, Y) = z) = \sum_{(x, y): g(x, y) = z} p_{X, Y}(x, y)$$

Expected value rule:

$$\rightarrow E[g(X, Y)] = \sum_x \sum_y g(x, y) p_{X, Y}(x, y)$$